

Full Stack Dart

Flutter Vikings - February 2022



A brief introduction



Engineer, The @ Company,
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Agenda

- Why Dart?
- JIT vs AOT
- Dart on Docker
- Functions Framework for Dart
- Profiling and performance management
- Back end architecture - x86 vs Arm trade offs
- Other places you can learn more
- Call to action - try out the Functions Framework Examples



Why Dart?



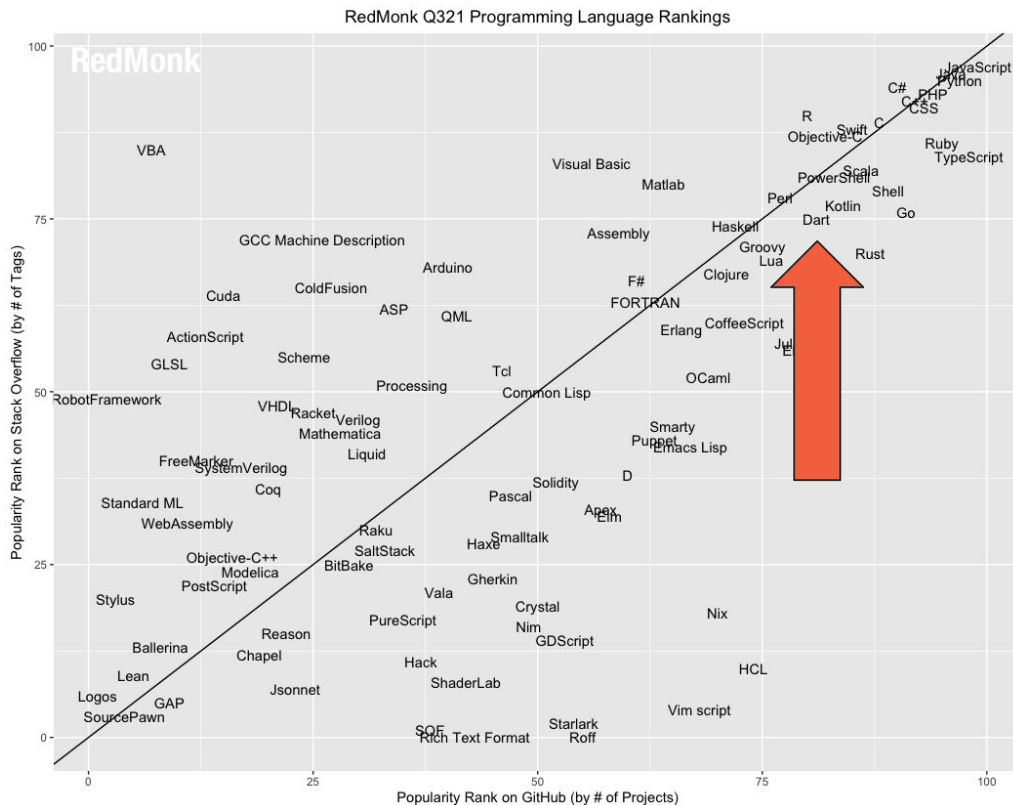
Dart

“Dart is a client-optimized language
for fast apps on any platform”

@dart.dev

Flutter pulled Dart into the Redmonk Top 20

- 1 JavaScript
- 2 Python
- 2 Java
- 4 PHP
- 5 CSS
- 5 C++
- 5 C#
- 8 TypeScript
- 9 Ruby
- 10 C
- 11 Swift
- 12 R
- 13 Objective-C
- 14 Shell
- 14 Scala
- 16 Go
- 17 PowerShell
- 18 Kotlin
- 19 Rust
- 20 Dart



If you're attending a
Flutter event then
you're probably
writing Dart already.
So why not use it for
the back end too?



Flutter



Dart

JIT vs AOT

Dartshowplatform - A more useful 'Hello World!'

```
import 'dart:io' show Platform, stdout;

void main() {
  print(Platform.version);
}
```

```
2.14.4 (stable) (Wed Oct 13 11:11:32
2021 +0200) on "linux_arm64"
```



JIT - Just `dart run` it in the virtual machine

```
$ time dart run showplatform.dart
```

```
2.14.4 (stable) (Wed Oct 13 11:11:32  
2021 +0200) on "linux_arm64"
```

```
real    0m6.280s  
user    0m5.050s  
sys     0m0.915s
```



AOT - Compile it first then run the binary

```
$ dart compile exe showplatform.dart  
$ time ./showplatform.exe
```

```
2.14.4 (stable) (Wed Oct 13 11:11:32  
2021 +0200) on "linux_arm64"
```

```
real    0m0.028s  
user    0m0.019s  
sys     0m0.009s
```



Trade off - compilation is slow

```
$ time dart compile exe \  
showplatform.dart
```

```
Info: Compiling with sound null safety  
Generated: showplatform.exe
```

```
real    0m17.434s  
user    0m20.912s  
sys     0m2.958s
```



Dart on Docker

A typical Dockerfile for AOT compiled Dart binary

```
FROM dart AS build
WORKDIR /app
COPY ./mydemoapp.dart .
RUN dart pub get
RUN dart compile exe /app/mydemoapp.dart \
    -o /app/mydemoapp

FROM scratch
COPY --from=build /runtime/ /
COPY --from=build /app/mydemoapp /app/mydemoapp
ENTRYPOINT ["/app/mydemoapp"]
```



Image sizes for a trivial application



atsigncompany/dartshowplatform ☆

By [atsigncompany](#) • Updated 6 days ago

Multi arch Dart test image that prints out version and platform info

Container

TAG

[latest](#)

`docker pull atsigncompany/dartshowpl...`



Last pushed 6 days ago by [atcpswan](#)

DIGEST

OS/ARCH

COMPRESSED SIZE ⓘ

[8f4d6b9d1d68](#)

linux/amd64

4.45 MB

[100829c90d10](#)

linux/arm/v7

3.26 MB

[1dac2840cb6e](#)

linux/arm64

3.68 MB



Image sizes for a non trivial application



atsigncompany/secondary ☆

By [atsigncompany](#) • Updated an hour ago

@protocol secondary server

Container

TAG

[dess](#)

docker pull atsigncompany/secondary:...



Last pushed 2 days ago by [atcpswan](#)

DIGEST

OS/ARCH

COMPRESSED SIZE ⓘ

[197123ec5e7a](#)

linux/amd64

5.97 MB

[1a75e98e541f](#)

linux/arm/v7

5 MB

[6b20a465d6cd](#)

linux/arm64

5.21 MB



Functions Framework for Dart

GoogleCloudPlatform/functions-framework-dart

The Functions Framework lets you write lightweight functions that run in many different environments, including:

- Your local development machine
- Google Cloud Run - see cloud run quickstart
- Google App Engine
- Knative-based environments

“Google Cloud Functions does not currently provide an officially supported Dart language runtime, but we're working to make running on Google Cloud Run as seamless and symmetric an experience as possible for your Dart Functions Framework projects.”



examples/hello/lib/functions.dart

```
import 'package:functions_framework/functions_framework.dart';  
import 'package:shelf/shelf.dart';  
  
@CloudFunction()  
Response function(Request request) => Response.ok('Hello, World!');
```

To:

```
$ curl https://"YOUR_APP_URL"  
Hello, World!
```

@ All without needing to worry about writing an HTTP server or request handling logic.

Profiling and performance management

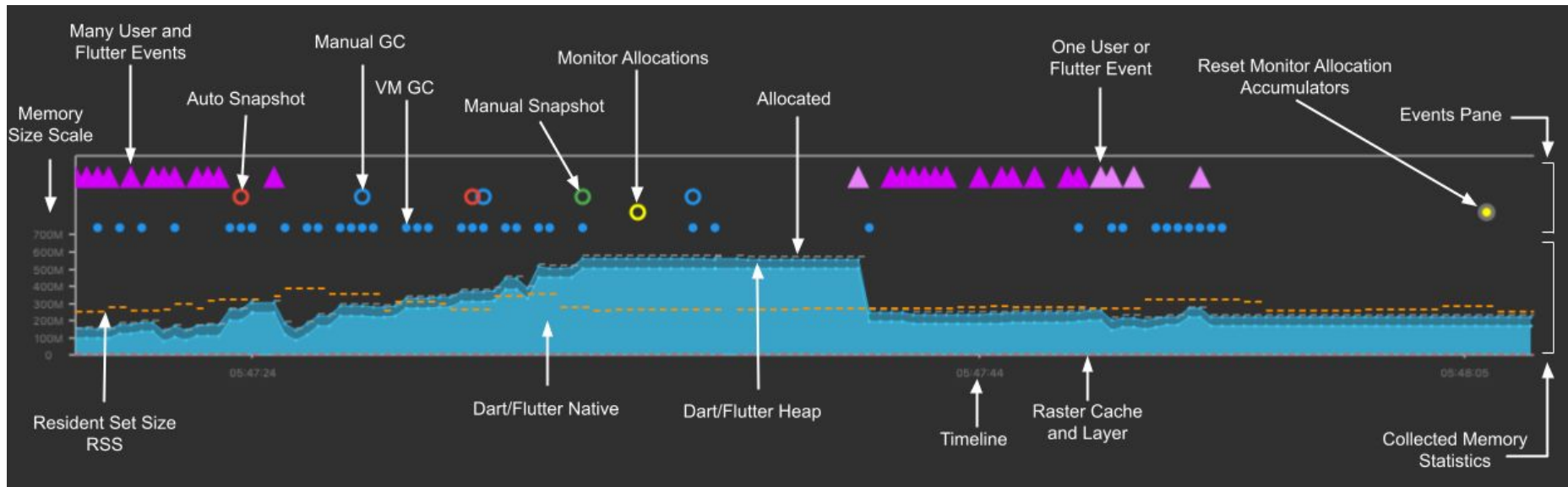
Dart DevTools

	Flutter mobile or desktop ↗	Flutter web ↗	Other web	Command-line
Debugger ↗	✓	✓	✓	✓
Logging view ↗	✓	✓	✓	✓
App size tool ↗	✓			✓
CPU profiler ↗	✓			✓
Memory view ↗	✓			✓
Network view ↗	✓			✓
Performance view ↗	✓			✓
Flutter inspector ↗	✓	✓		

<https://dart.dev/tools/dart-devtools>



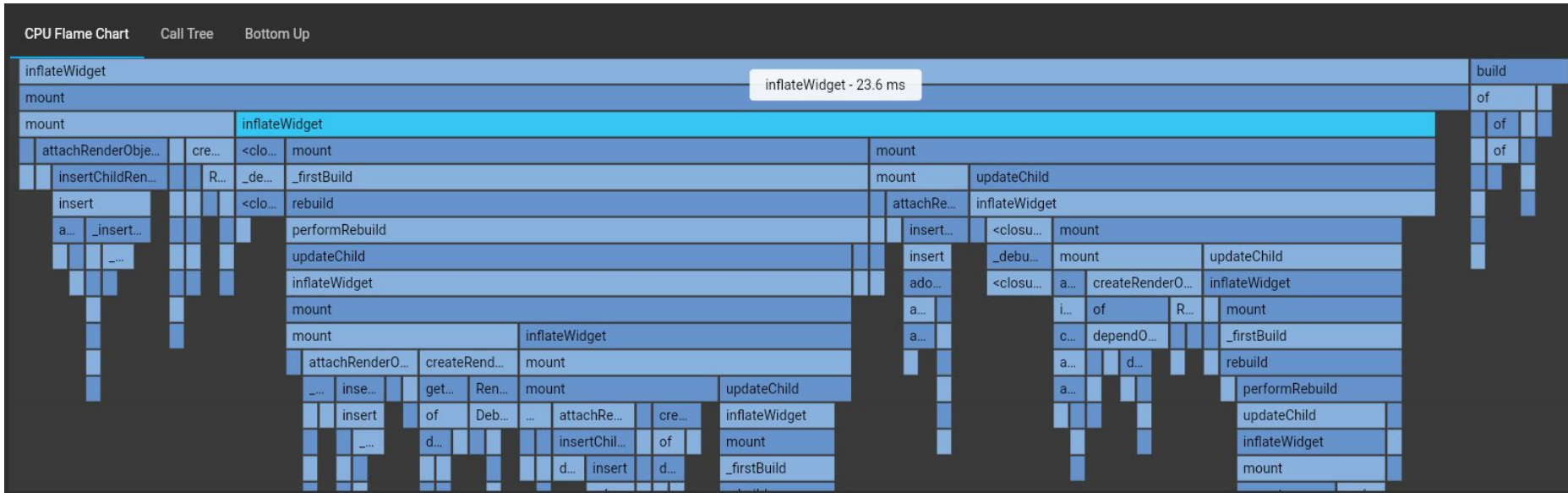
Memory view



<https://flutter.dev/docs/development/tools/devtools/memory>



Flame chart



<https://flutter.dev/docs/development/tools/devtools/performance>



But... DevTools need to connect to a VM, so JIT only

[@mraleph](#) where would you start with trying to track down a potential memory leak in an AOT Dart app (running in a container)?

I'm presently planning to run an instance of it in a JIT VM so I can use the memory profiler, but if there's a way to get metrics from an AOT binary?

6:20 pm · 12 Aug 2021 · Twitter Web App



Vyacheslav Egorov @mraleph · 12 Aug



Replying to [@cpswan](#)

yeah, this is a bit complicated - if it was a Flutter application you could use so called profile mode (AOT with profiling tools enable) to collect a heap profile. but dart compile exe always produces release mode snapshots.



1



1



Vyacheslav Egorov @mraleph · 12 Aug



if you feel adventurous you could build your own SDK and change dart compile to produce vmervice enabled binaries.



Back end architecture

Dart SDK Architectures

OS	Architecture
Linux	x64
	ia32
	ARMv8 (ARM64)
	ARMv7
macOS	x64
	ARM64
Windows	x64
	ia32



Readily Available in the Cloud

OS	Architecture
Linux	x64
	ARMv8 (ARM64)



Common IoT platforms

OS	Architecture
Linux	ARMv8 (ARM64)
	ARMv7



Docker BuildX *can* build ARM images on x64

So it's possible to generate multi platform images from hosted CI/CD pipelines such as GitHub Actions.

But...

- Uses QEMU, so can be slow, which can seriously impact CI pipeline commit cycles
- Pretty reliable for ARMv8, not so much for ARMv7



Multi Platform Images Without Using BuildX

Each architecture needs to be built on its own underlying hardware.

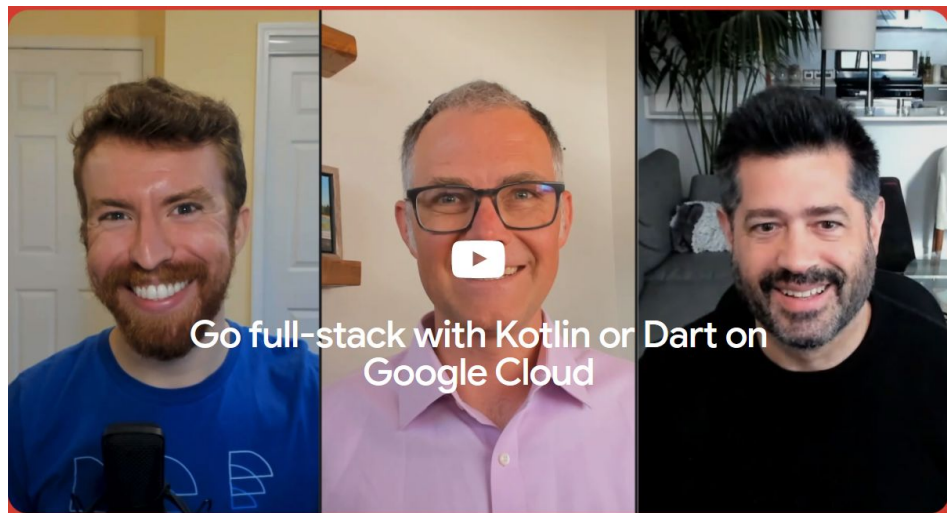
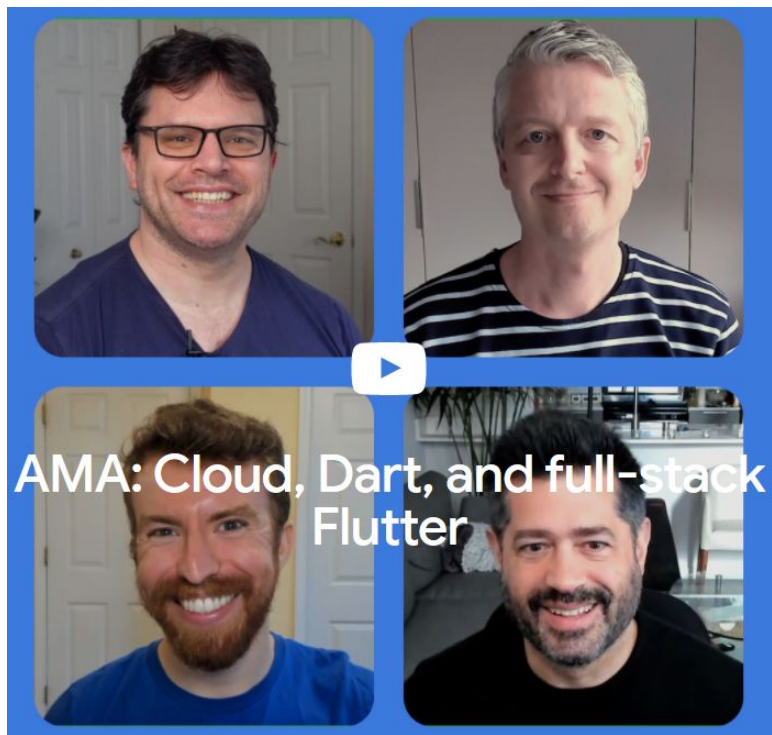
- X64 in regular CI
- AWS Graviton instances for ARMv8
- Raspberry Pis for ARMv7

Then a multi platform manifest can be assembled from the platform specific images.



Other places where you can learn more

<https://events.google.com/io/session/083b441e-4076-4443-9750-6529b34257b7?lng=en>



<https://events.google.com/io/session/6d5795f2-ec1c-4c7a-9efa-ea90c8c59c03?lng=en>



Full Stack Dart

with Tony Pujals and Kevin Moore

<https://gcppodcast.com/post/episode-261-full-stack-dart-with-tony-pujals-and-kevin-moore/>

<https://pinboard.in/u:cpswan/t:dart>

★ [yaml_edit | Dart Package](#) ✓

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★ [Experimenting with compiling Dart to Wasm, and calling Wasm from Dart](#) ✓

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★ [Iakos - Visualize Dart library dependencies in Graphviz](#) ✓

[dart](#)

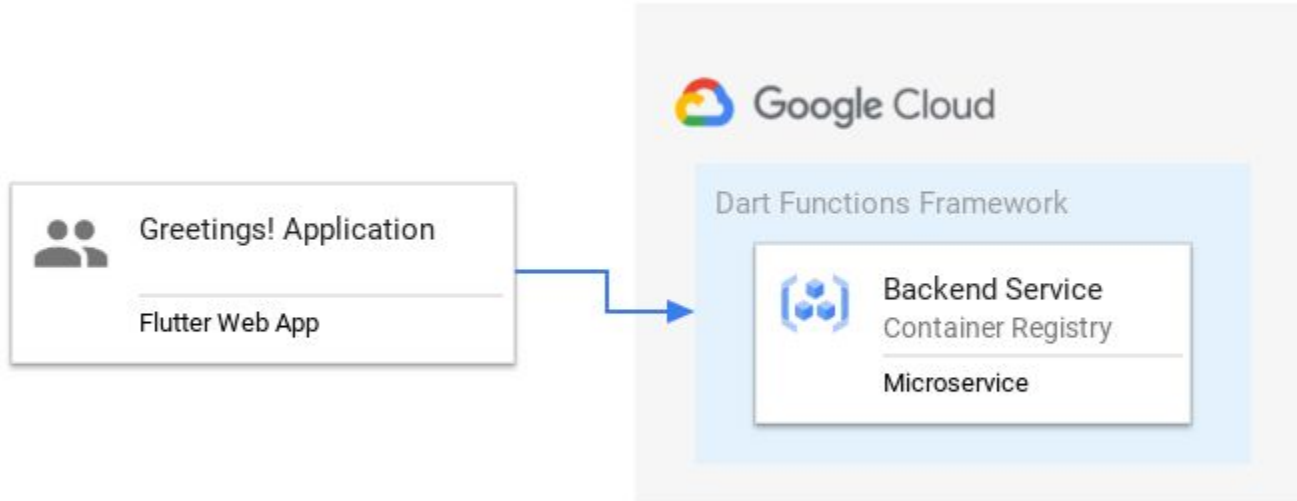
july 2021 by [cpswan](#) [edit](#) [delete](#)



Call to action:
Try the Functions Framework Examples

Qwiklabs - Dart Functions Framework

Dart Functions Framework



Free Qwiklabs from Cloud Skills Challenge

Start your skills challenge today

Choose a challenge track and race the clock to earn your recommended skill badges. Be sure to complete the challenge in 30 days with your free access. Unlock a new challenge every month.

Start challenge



<https://inthecloud.withgoogle.com/google-cloud-skills/register.html>

Review

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Contact

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The @ Company



Get started with the @platfrom



<https://atsign.dev/>



<https://github.com/atsign-foundation>



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Questions?